LEAKLISTEN · PRODUCT REQUIREMENTS AND SPECIFICATIONS

LeakListen product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-11-01	Amish Chadha	Converted to the kit 1.9.1 product requirements and specifications format (LKL-REQ-001 becomes LKL-PRS-001): R1 to R15 become PR-001 to PR-015 and S1 to S4 become PR-016 to PR-019, each traced to a user requirement in LKL-URS-001 and listing its old number; PR-020 (legal transmit power) added from LKL-CAL-001 D4; the status of each requirement is unchanged; mass re-run on the D3 model (1.34 kg with the neck bar)
0.7	2026-10-02	Amish Chadha	R6 figure and R15 cost updated for the status light pipe (LKL-DDR-003 follow-up); no status changed
0.6	2026-10-02	Amish Chadha	R12 restated by Amish on 2026-10-02 (LKL-DDR-003 A1): 1.0 kg for the logger, cable and sensor, neck bar counted as site hardware; met on paper
0.5	2026-10-01	Amish Chadha	

			Constructable design (LKL-DDR-003): status from LKL-CAL-001 v0.3; R12 not met on mass with the neck bar; R15 reported against the value-engineering target; R3 and R10 figures updated
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002): R1 and R2 restated (contact sensor for metallic mains, hydrophone variant for plastic); R7 restated against a district gateway within 0.5 km under iron covers; R9 met with the 42 mm magnet; status from LKL-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3, status from LKL-CAL-001 for every requirement; R8 fixed (LKL-DDR-001 D6); R5 and R14 note the weekly time correction and the 24-byte summary
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

LeakListen product requirements and specifications

These are the product requirements for one logger, each traced to a user need in LKL-URS-001 (links in LKL-RTM-001); they replace LKL-REQ-001, whose numbers R1 to R15 and S1 to S4 are kept in the "Was" column. The requirements describe option A, the current design, which has not been through a concept choice yet (LKL-OPT-001 recommends it; "Proposed, awaiting Amish"). Targets are proposals for review; they come from desk research, not yet from a partner utility, and must be revised after co-design (see the problem statement, LKL-PRB-001). Status is judged against the design in LKL-PRC-001 v0.5 by the calculations in LKL-CAL-001 v0.2, which print every figure quoted here. The TRL 2 review decisions are recorded in LKL-DDR-001. The TRL 3 review decisions, accepted by Amish on 2026-09-25, are recorded in LKL-DDR-002: they restate R2 (plastic mains are served by the hydrophone variant) and R7 (a gateway within 0.5 km of each district with iron covers); no numeric target is lowered.

Table 1. Product requirements and specifications. Status at TRL 3 from LKL-CAL-001 (python docs/04-calcs/sizing.py).

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Performance	Hears a leak on metallic mains	Flag a 5 L/min (1.3 US gal/min) leak at 3 bar within 100 m of the logger on cast or ductile iron DN100 to DN200. The magnet-on contact sensor's scope is metallic mains (LKL-DDR-002)	100	max distance	m	Attenuation and detection estimate (LKL-CAL-001 section G); later a partner site with a controlled leak	**At risk:** about 185 m on ductile iron at the central estimate (signal equal to noise), about 140 m if the leak must lift the level by the 6 dB of the flag rule (independent review, LKL-STM-001 S2); 93 to 370 m over plausible pipe losses	R1
PR-002	UR-002	Performance	Hears a leak on plastic mains (hydrophone variant)	Same leak within 30 m on PVC or PE DN100 to DN200, with the hydrophone variant on a hydrant or tapping point (LKL-DDR-002, D3; option B in LKL-OPT-001)	30	max distance	m	As PR-001, for the hydrophone variant	Open: the hydrophone variant is not sized in LKL-CAL-001. First-order screen only (LKL-STM-001 SB1): 54 m on PVC at an assumed 0.1 Pa/rtHz background, none at 1 Pa/rtHz. For the record the contact sensor hears about 4 m on PVC and 2 m on PE	R2
PR-003	UR-001	Performance	Sensor bandwidth	5 Hz to 2 kHz, within ±3 dB after correction	5 to 2,000	±3 dB	Hz	Resonance estimate (LKL-CAL-001 section F);	**At risk:** with the 42 mm magnet and the 8 mm puck base the mount	R3

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
								later a shaker comparison	resonates at about 676 to 1,350 Hz, flat only to 370 to 730 Hz	
PR-004	UR-001	Performance	Sensor self-noise	Equivalent input noise 1 µg/√Hz or less from 100 Hz to 1 kHz	1	max	µg/√Hz	Noise calculation (LKL-CAL-001 section F)	**At risk:** 1.03 µg/√Hz worst case with the 53 g seismic mass	R4
PR-005	UR-005	Performance	Night listening	12 windows of 20 s between 02:00 and 04:00 local time, clock drift under 1 min per month	60	max drift per month	s	Clock drift estimate (LKL-CAL-001 section H)	Met on paper with a weekly network time correction (20 s worst); 87 s a month free running	R5
PR-006	UR-007	Electrical	Battery life	5 years or more on one primary cell at the PR-005 schedule and one uplink a night	5	min	years	Power budget (LKL-CAL-001 section C)	Met on paper: 10.1 years or more on energy; taken as 10 years	R6
PR-007	UR-005	Electrical	Radio link	90 % or more of nightly summaries delivered from a chamber with a cast-iron cover to a gateway planned within 0.5 km of each such district; 1 km where the utility agrees to a composite cover or a through-cover antenna (LKL-DDR-002)	90 at 0.5	min	% at km	Link budget (LKL-CAL-001 section D); cover loss measurement and field check later (TRL 4, on hold)	**At risk:** +0.9 dB at 0.5 km under a 20 dB iron cover (EU868 SF12), -1.2 dB for US915 SF9, -9.1 dB for a tight 30 dB cover; cover loss unmeasured	R7
PR-008	UR-009	Software	Data and privacy	Only band levels and spectra leave the logger; raw vibration	Pass or fail	Pass or fail	n/a	Design review	Met by design	R8

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				samples are deleted on the device after processing. Fixed requirement (LKL-DDR-001 D6)						
PR-009	UR-003, UR-012	Physical architecture	Attachment	Magnet holds 100 N or more on a steel or iron spindle cap, with no tools or pipe work	100	min	N	Hold estimate (LKL-CAL-001 section I); a pull test later	Met on paper with the 42 mm magnet (LKL-DDR-002): 125 N on a 0.5 mm coating, 187 N on 0.3 mm. Non-ferrous fittings, outside this target, need an adapter	R9
PR-010	UR-003, UR-004	Usability	Install from the surface	Placed and recovered in 10 min or less without entering the chamber, for chambers up to 1.5 m deep	10	max	min	Task estimate (LKL-CAL-001 section K); a walk-through with the partner utility	Not verifiable at TRL 3; the 11 min estimate (neck bar assembled beforehand) is 1 min over the target, so it is at risk on the estimate; cable reaches caps 2.18 m down	R10
PR-011	UR-006	Physical architecture	Survive the chamber	IP68: 1 m submersion for 7 days; operate from -20 to +50 °C	IP68; -20 to +50	Pass or fail	class; °C	Design review (LKL-CAL-001 section J)	Met by design; the logger floats on its lanyard when flooded	R11
PR-012	UR-014	Packaging and transport	Size and mass	Logger 70 mm diameter or less and 300 mm long or less; logger, cable and sensor 1.0	70 x 300; 1.0	max	mm; kg	Parametric model (LKL-CAL-001 section K); weighing at TRL 4	Met on paper: 68 x 283 mm overall; logger, cable and sensor 0.96 kg (0.04	R12

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				kg or less. The neck bar is site hardware that stays in the chamber and is not counted (restated by Amish on 2026-10-02, LKL-DDR-003 A1)					kg margin); the neck bar is 0.31 kg. The 70 g antenna rides on the bar and is counted with it; with the antenna the carried set is 1.03 kg (independent review 2026-11-01): the exclusion is Proposed, awaiting Amish to confirm	
PR-013	UR-013	Performance	Leak flag quality	1 false alarm or fewer per 20 loggers per month after a 14-night baseline	1 per 20	max	per month	Field data from a partner network	Not verifiable at TRL 3	R13
PR-014	UR-010, UR-015	Software	Open and interoperable	Standard LoRaWAN 1.0.x uplink (1.0.3 for the time request), documented 24-byte payload, works with any network server and TwinKit	24	max payload	B	Design review (LKL-CAL-001 sections A and B)	Met by design	R14
PR-015	UR-008	Cost	Cost	Parts cost per logger against a value-engineering target of USD 130 (`budget_usd` , a hypothetical control target)	130	max	USD	Priced BOM (LKL-CAL-001 section M)	USD 142, USD 12 over the value-engineering target	R15
PR-016		Safety					n/a			S1

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
	UR-003, UR-011		No confined space entry	No confined space entry is needed to place, service or recover a logger	Pass or fail	Pass or fail		Design review; build plan steps (LKL-BLD-001)	Met by design (surface placement by pole, PR-010)	
PR-017	UR-011	Safety	Safe cell	The cell is a primary lithium cell with a fuse or PTC and reverse protection; no charging circuit is fitted	Pass or fail	Pass or fail	n/a	Design review; main board design at TRL 4	Met by design on paper; the board is not designed at TRL 3	S2
PR-018	UR-011, UR-016	Safety	Labelled	The logger carries a label with the owner, a contact and a lithium cell warning	Pass or fail	Pass or fail	n/a	Inspection of BOM line 11 and the build plan	Met by design (label in BOM line 11)	S3
PR-019	UR-012	Safety	Nothing touches the water	Nothing touches drinking water, and nothing is fixed to the pipe other than the magnet	Pass or fail	Pass or fail	n/a	Design review of the model (no part passes through a pipe wall)	Met by design; applies to option A only (option B's hydrophone is in the water, LKL-REG-001 G4)	S4
PR-020	UR-016	Regulatory	Legal transmit power	Transmit power within the regional limit: +14 dBm ERP in EU868 (16.15 dBm EIRP); up to +20 dBm in US915 where permitted	14	max	dBm	Design review (LKL-CAL-001 D4); regulatory check at TRL 4 (LKL-REG-001)	Met by design in EU868 at +14 dBm; the TRL 2 figure of +20 dBm conducted is not permitted there unless the antenna gain and lead loss bring it under 14 dBm ERP (independent review): the free-field antenna gain is to assess	(LKL-CAL-0 D4)

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software, cost, regulatory. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Requirements at risk or open

The notes below keep the old numbers R1 to R15; R-n is PR-0n (R1 is PR-001, R15 is PR-015).

Summary (LKL-CAL-001 v0.6, Table 6): none not met, 4 at risk, 1 open (R2, variant not sized), 2 not verifiable at TRL 3, 4 met on paper, 3 met by design, and R15 USD 12 over its value-engineering target. R12 was not met on mass after LKL-DDR-003 until Amish restated it on 2026-10-02 (A1); before LKL-DDR-003 R12 and R15 were met on paper; before LKL-DDR-002 there were 2 not met (R2, R7) and 4 at risk (R1, R3, R4, R9).

- **R12 thin margin on mass.** The concept's strap hanger could not be built (it had nothing to hook over except the seat under the cover), and the neck bar that replaces it (LKL-DDR-003) weighs 0.31 kg (0.32 kg before the D3 rebuild). The set is 1.34 kg; the logger, cable and sensor are 0.96 kg. Amish decided on 2026-10-02 (LKL-DDR-003, A1) to restate R12 as 1.0 kg for the logger, cable and sensor and count the bar as site hardware, so R12 is met on paper with a 0.04 kg margin; the logger set is weighed at TRL 4.
- **R7 at risk.** At an assumed 20 dB cover loss the link closes to about 0.53 km in EU868 and 0.46 km in US915. Under LKL-DDR-002 a gateway is planned within 0.5 km of each district with iron covers, which EU868 SF12 reaches with +0.9 dB to spare and US915 SF9 misses by 1.2 dB. The cover loss must be measured first; that is TRL 4 work and on hold.
- **R2 open.** Leak noise attenuates quickly on plastic pipes (0.42 dB/m at 100 Hz on PVC against 0.011 dB/m on ductile iron), and a contact sensor on a valve hears the reference leak only a few metres along PVC or PE. Under LKL-DDR-002 R2 applies to the hydrophone variant (LKL-DDR-001 D3), which is not yet sized.
- **R1 at risk.** The central estimate of 185 m on iron clears 100 m, but pipe loss and leak source strength are both uncertain by large factors.
- **R3 at risk.** The magnet mount's resonance falls inside the band and varies by site; the heavier 42 mm magnet and the 8 mm puck base lower it by about 13 %.
- **R4 at risk.** 1.03 µg/√Hz against 1 µg/√Hz, on assumed ceramic and op-amp data.
- **R9 met on paper** with the 42 mm magnet (LKL-DDR-002); it was at risk with the 32 mm magnet (90 N at a 0.3 mm coating). Brass, bronze or plastic fittings still need an adapter.
- **R10 and R13 not verifiable at TRL 3.** They need a walk-through and field data with a partner utility.

Assumptions

- Network pressure at night 2 to 6 bar; leaks at lower pressure are quieter.
- Valve spindle caps and hydrant bodies are cast iron or steel on most networks the lab would target first.
- A LoRaWAN gateway (TwinKit, a partner's or a public network) is within about 1 km in an urban area, and within about 0.5 km of each district with cast-iron covers (LKL-DDR-002).
- Transmit power follows the regional limit: +14 dBm in EU868, up to +20 dBm in US915.
- Primary cell: lithium thionyl chloride C size, 3.6 V, about 7.7 Ah nominal, derated to 70 % usable for cold and pulse loads.

Safety requirements

Safety: PR-016 (was S1): no confined space entry is needed to place, service or recover a logger. PR-017 (was S2): the cell is a primary lithium cell with a fuse or PTC and reverse protection; no charging circuit is fitted. PR-018 (was S3): the logger carries a label with the owner, a contact and a lithium cell warning. PR-019 (was S4): nothing touches drinking water, and nothing is fixed to the pipe other than the magnet.